GERASIMOS BALATSOUKAS

Mechanical and Biomedical Design Engineer | SolidWorks, Fusion 360, Abaqus, Ansys
+44 7565 147362 balatsoukas.gerasimos@gmail.com LinkedIn Portfolio
Full UK right to work (EU Settlement Scheme) | EU citizen | Available now | Driving licence | Will relocate

EDUCATION

Cranfield University **Cranfield, UK | Sep 2025 – Sep 2026**
MSc in Advanced Mechanical Engineering (Distinction)
Thesis: Effect of SO_2 Concentration and Alkali Deposition on the Corrosion Behaviour of Supercritical CO_2 Turbine Blade Alloys (in collaboration with Siemens Energy)

University of Glasgow **Glasgow, UK | Sep 2020 – May 2024**
BEng Hons in Biomedical Engineering (Upper Second Class)
Thesis: Construction of 3D Piezoelectric Scaffolds for Muscle Regeneration, *68%*
Publication: Second author, "Three-Dimensional Piezoelectric Fibrous Constructs for Muscle Regeneration", *Small* (2026), doi:10.1002/smll.76613

RESEARCH & PROJECT EXPERIENCE

University of Glasgow, Advanced Research Centre **Glasgow, UK | Sep 2023 – Apr 2024**
Undergraduate Researcher: Piezoelectric Scaffolds for Muscle Regeneration
- Traced tensile readings orders of magnitude off the literature to the load-cell calibration in the rig's Arduino controller, re-characterised it in C++ and validated it against a calibrated rig
- Designed and 3D printed new rig components, reorienting the rig 90 degrees for submerged wet testing
- Fabricated electrospun scaffolds that mimic muscle and modified their surface polarity to improve cell adhesion, then assessed biocompatibility and differentiation

Personal Project: Coronary Stent Design and Expansion Model **2025**
- Built a parametric stent model in Fusion 360 and simulated balloon expansion and springback in Abaqus with nonlinear plasticity, reaching mesh independence within 5% under a 250,000-node licence limit

Cranfield University **Cranfield, UK | 2026**
MSc Thesis Researcher: High-Temperature Corrosion (Siemens Energy collaboration)
- Ran a 1,000-hour furnace test on nickel- and cobalt-base alloys in a simulated supercritical CO_2 environment, controlling SO_2 concentration and salt deposition
- Prepared metallographic cross-sections, measured metal loss by optical metrology and identified corrosion mechanisms with SEM and EDX

Cranfield University **Cranfield, UK | 2025 – 2026**
Group Project: Rare-Earth-Free Wind Turbine Generator (Team of Four)
- Simulated a 2 MW wound-rotor generator in Ansys Maxwell and RMxprt, reaching 2.5 MW peak at 92.7% efficiency

PROFESSIONAL EXPERIENCE

PwC **Nicosia, Cyprus | Jun 2025 – Aug 2025**
Business Consulting Summer Intern
- Worked on two semi-governmental proposals (healthcare and telecoms), each over €200,000, both won

Hellenic Army: Compulsory National Service, Artillery **Greece | Sep 2024 – Jun 2025**

Fender S.A. (Cambiaso Risso Group) **Athens, Greece | Jun 2023 – Aug 2023**
Data Management Summer Intern: Marine Insurance
- Consolidated multinational contract data into a proprietary system, cutting contract generation time by 50%

Agency for Quality Assurance, Greek Ministry of Health **Athens, Greece | Jun 2022 – Aug 2022**
Summer Intern: Hospital Quality and Safety
- Checked hospital defibrillators and equipment for emergency readiness in Thessaloniki and Chania (ISO 13485)
- Drafted directives for the "Safe Hospitals" project, extending emergency SOPs to all Greek public hospitals

ADDITIONAL SKILLS

CAD and Prototyping: SolidWorks, Autodesk Fusion 360, 3D/Extrusion Printing (all Intermediate)
Simulation: Abaqus FEA, Ansys Fluent and ICEM CFD, Ansys Maxwell (Intermediate), Simulink (On-Ramp Certified), COMSOL, MoFEM (Beginner)
Coding: Python (Advanced), MATLAB (Advanced), HTML5 and CSS (Intermediate), R (Beginner)
Laboratory: SEM, Optical Microscopy, Cell Culture (Advanced), qPCR (Intermediate), Metallography, EDX Mapping, Electrospinning
Languages: English (Native), Greek (Native), German (Intermediate, B1 & IB Certification)
Interests: Kickboxing and Brazilian Jiu-Jitsu, motorcycle maintenance and restoration